SEARCHING FOR BLACK-HOLE STARS

Biswajit Jana | JWST Cosmic Anomaly Lab | Research progress: 12 November 2026

CONFERENCE POSTER 01

CEERS / JADES / RUBIES / UNCOVER

Public observations. Reproducible analysis.

No new black-hole-star identification.

01 / THE RESEARCH QUESTION

Can unusual JWST sources be explained by stars, ordinary or obscured active galactic nuclei, dense accretion envelopes, or a mixture of these? The aim is to distinguish these possibilities with observations and competing predictions.

02 / A DEFINED DATASET

2,647 / 2,647 eligible source records analysed.
PRISM spectra; archive grade 3; median pixel S/N at least 5; 0.5 <= z < 12. One highest-S/N observation per upstream source ID. No colour or physical-class filter.

This is 13.1% of the parent catalogue's 20,144 source records, not a complete JWST census. Targeting, depth and redshift success affect selection.

03 / FROM ARCHIVE TO EVIDENCE

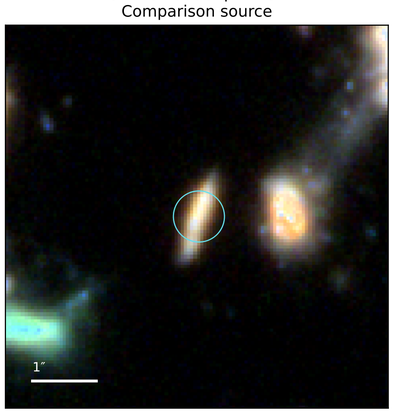
1. Verify units, quality masks and product checksums.
2. Measure continuum windows and empirical line blends.
3. Vary instrument resolution and assumed errors.
4. Inspect images, 2D spectra and higher-resolution data.
5. Check published benchmarks and competing explanations.

11 follow-up priorities.
A screening pass is not a physical identification.

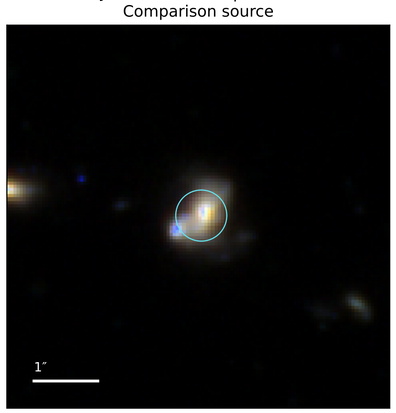
ACTUAL TELESCOPE DATA

ACTUAL NIRCAM IMAGES · FALSE COLOUR
F444W → red F277W → green F150W → blue | 1″ scale bars
Colour and apparent compactness do not establish a black-hole star. PSFs unmatched.

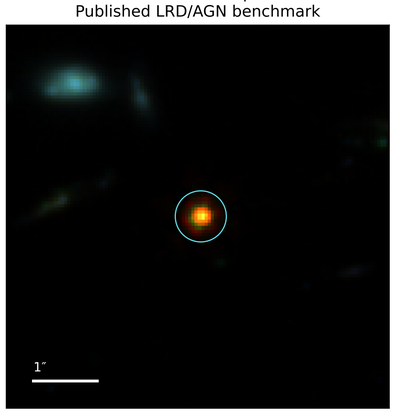
CEERS · 5336 | z 1.54
Comparison source



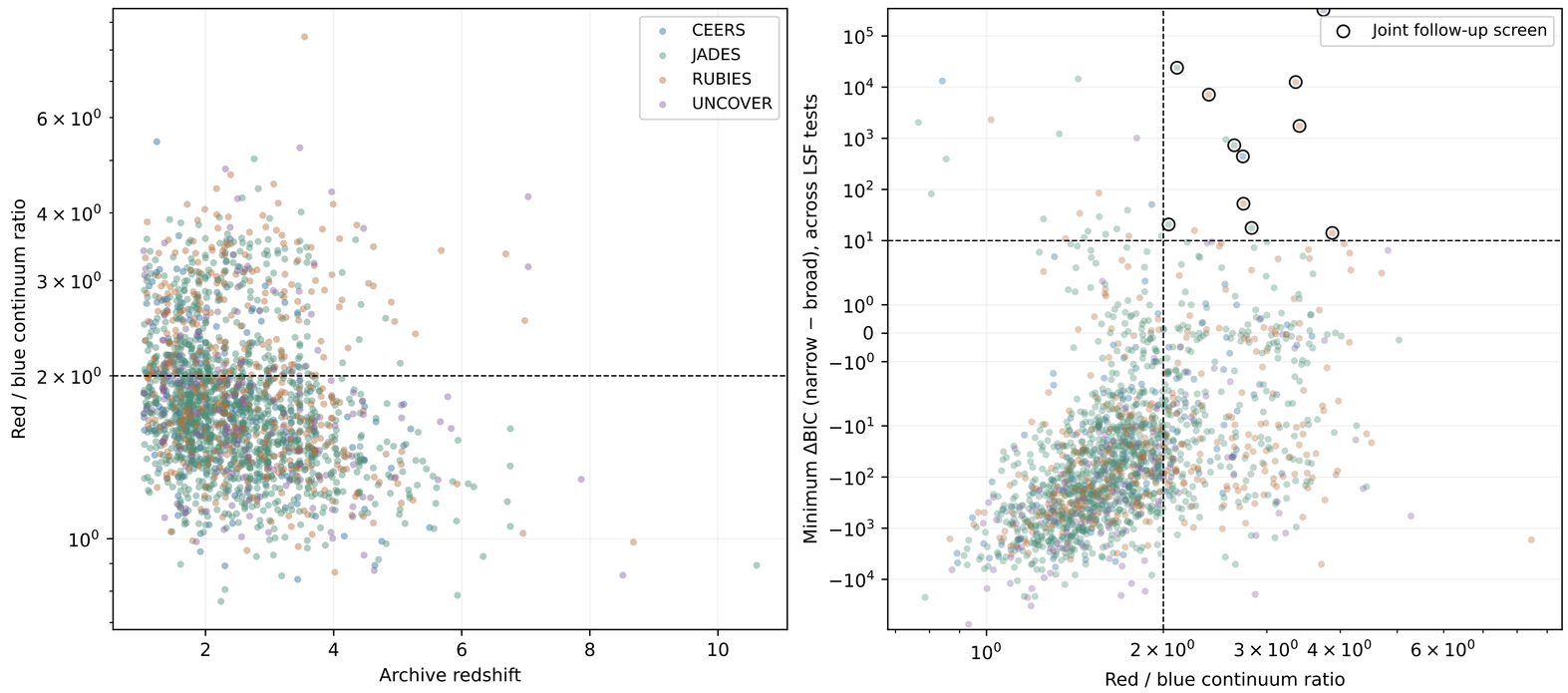
JADES · 201011 | z 1.76
Comparison source



UNCOVER · 45924 | z 4.47
Published LRD/AGN benchmark



OBSERVED · 2,647 spectra screened | 11 follow-up priorities
Descriptive thresholds; neither a physical classification nor a calibrated discovery significance



Data credit: NASA / ESA / CSA / STScI; CEERS, JADES, RUBIES, UNCOVER; DAWN JWST Archive / msaexp. Images are false-colour NIRCam cutouts; PSFs are unmatched. Full methods, data hashes and code: github.com/Biswajit1999/JWST-Cosmic-Anomaly-Lab